

Do First Generics Reach Medicaid Pharmacy Markets Equally After Approval? An ML.NET Framework for Measuring Diffusion and Access Gaps

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Abstract

Regulatory approval of a first generic drug is often treated as a meaningful access milestone because it permits generic entry for a previously brand-only product. However, approval does not guarantee that utilization appears immediately or broadly across payer markets. This study develops and evaluates an end-to-end, reproducible SQL Server and ML.NET workflow for measuring and forecasting post-approval diffusion of first generic drugs across state-level Medicaid pharmacy markets. FDA first-generic approval records from 2017–2024 were linked with Medicaid State Drug Utilization Data from 2017–2025. The SQL Server pipeline stages FDA approval records and annual Medicaid utilization files, deduplicates launch records, maps generic and brand names to Medicaid utilization records, constructs quarterly launch-level diffusion metrics, exports model-ready datasets, and produces final result tables. Two complementary distribution measures are defined: Numeric Distribution, the percentage of active state-level Medicaid molecule markets where generic utilization is observed, and Weighted Distribution, the share of molecule-level prescription volume represented by those generic-observed states. Their difference, Access Gap, identifies high-volume-first diffusion, where generic utilization appears in large Medicaid markets before becoming geographically broad.

After deduplication, the final analysis dataset contained 278 included first-generic launch records, 3,202 quarterly diffusion observations across quarters 1–13 after approval, 1,064 first-four-quarter observations, and 265 complete first-year launch cases. Among complete Q1–Q4 cases, 202 launches were classified as Fast adopters, 6 as Medium adopters, and 57 as Slow adopters based on fourth-quarter Weighted Distribution. A year-based ML.NET multiclass model trained on 2017–2022 launch observations and tested on 2023–2024 observations achieved 90.78% micro accuracy, 65.08% macro accuracy, and log loss of 0.4734. The model separated Fast and Slow adoption patterns well, with F1-scores of 92.49% and 96.00%, respectively, but did not identify the rare Medium class. The results show that early Medicaid diffusion signals can support first-year adoption monitoring, especially for distinguishing broad versus weak diffusion, while intermediate adoption remains difficult to model because of class imbalance and threshold sensitivity. The main contribution is a transparent, reproducible public-data framework for moving from regulatory approval records to interpretable diffusion metrics and early adoption forecasting.

Keywords: generic drugs; Medicaid; State Drug Utilization Data; first generic approval; drug diffusion; access gap; Numeric Distribution; Weighted Distribution; ML.NET; SQL Server; pharmaceutical analytics

1. Introduction

The approval of a first generic drug is an important event in the pharmaceutical lifecycle. It indicates that a generic version of a previously brand-only medicine has received regulatory approval and may enter the

market. Because generic competition can reduce costs and improve affordability, first generic approval is often viewed as a signal that access to lower-cost therapy should begin improving [6,7].

However, regulatory approval is not the same as realized utilization. A generic product may be approved but may not immediately appear in payer utilization data. Even after utilization begins, adoption may not be geographically uniform. A generic can appear first in a small number of large states, generating substantial prescription volume while remaining absent from many smaller markets. Conversely, a product can appear across many states but still account for a modest share of prescription volume. These differences matter because approval-based measures alone cannot show whether the access benefit of generic entry is broad, delayed, concentrated, or uneven.

This study examines the post-approval diffusion of first generic drugs in state-level Medicaid pharmacy markets. The core idea is that first generic approval should be treated as the starting point of a measurable diffusion process rather than as the final proof of access. Medicaid State Drug Utilization Data provide repeated state-quarter utilization records, making it possible to observe whether generic utilization appears after approval and how diffusion changes over time.

The study introduces two complementary measures. Numeric Distribution measures geographic breadth: the share of active state-level molecule markets where generic utilization is observed. Weighted Distribution measures volume-weighted reach: the share of molecule prescription volume represented by states where generic utilization is observed. The difference between them, defined as Access Gap, measures whether generic utilization is concentrated in high-volume markets before becoming geographically broad.

The study also evaluates whether early diffusion patterns can forecast first-year adoption. An ML.NET multiclass classification model is trained using early post-approval features from quarters 1 through 3 and evaluated against fourth-quarter adoption labels. The main validation uses a year-based split: launches from 2017–2022 are used for training, while launches from 2023–2024 are used for testing. This design approximates a prospective setting in which historical launch behavior is used to predict newer launch outcomes.

1.1 Research Questions

This study is guided by four research questions.

RQ1. After FDA first-generic approval, how quickly does generic utilization appear across state-level Medicaid pharmacy markets?

RQ2. Do Numeric Distribution and Weighted Distribution reveal different patterns of post-approval diffusion, particularly cases where utilization is concentrated in high-volume states before becoming geographically broad?

RQ3. Which first generic launches exhibit the largest Access Gap between volume-weighted reach and geographic breadth?

RQ4. Can early post-approval diffusion signals from the first three quarters predict whether a launch will achieve broad weighted Medicaid adoption by the fourth quarter?

1.2 Contributions

This paper makes five contributions.

First, it presents a reproducible SQL Server workflow that links FDA first-generic approval records with Medicaid State Drug Utilization Data and converts them into launch-quarter diffusion metrics.

Second, it adapts the concepts of Numeric Distribution and Weighted Distribution to state-level Medicaid generic diffusion, allowing geographic breadth and volume-weighted reach to be measured separately.

Third, it defines Access Gap as a practical indicator of high-volume-first diffusion, identifying cases where generic use appears in large markets before becoming geographically broad.

Fourth, it implements an ML.NET forecasting workflow that uses early diffusion features to predict fourth-quarter adoption class.

Fifth, it reports a transparent year-based validation, showing strong Fast/Slow separation but weak Medium-class performance due to rare intermediate adoption cases.

2. Background

Generic drug approval is commonly associated with improved affordability and competition. A first generic approval is especially important because it marks the first approved generic version of a brand product. Nevertheless, the practical effect of approval depends on whether the product is marketed, distributed, covered, prescribed, dispensed, and ultimately observed in utilization data.

Generic diffusion can vary for many reasons. Supply availability, therapeutic complexity, pharmacy purchasing, payer coverage, substitution rules, prior authorization, prescriber behavior, state Medicaid policies, and product-specific factors may all influence the speed and breadth of adoption. Prior work has shown that uptake of new medicines can be influenced by policy, market, clinical, and organizational factors [8]. Public approval records alone cannot reveal these dynamics.

Medicaid utilization data provide an opportunity to study diffusion using public administrative records. Medicaid State Drug Utilization Data are reported by state and quarter for covered outpatient drugs. Medicaid claims and utilization data are useful for population-level drug-use research, but they have known limitations for patient-level inference [9]. Although these data do not represent the entire U.S. prescription market, they provide a consistent public source for observing state-level utilization patterns over time.

Most simple adoption measures are insufficient because they collapse different forms of diffusion into one number. A total prescription count can be high even if only a few large states have adopted the generic. A state count can be high even if generic use is concentrated in low-volume states. For this reason, this study separates breadth from weighted reach. Numeric Distribution captures the share of active state markets showing generic use. Weighted Distribution captures the share of molecule prescription volume represented by those states. Access Gap captures the imbalance between the two.

This measurement design is especially useful for first generic launches because the earliest quarters after approval are analytically important. They reveal whether utilization is immediate, delayed, concentrated, broad, or absent. These early signals can also be used as predictive features for later adoption status.

3. End-to-End System Pipeline

A central contribution of this study is the reproducible workflow that converts public regulatory and utilization data into diffusion metrics, model-ready datasets, validation outputs, and result tables. The workflow was implemented using SQL Server for data engineering and ML.NET for machine learning.

The pipeline begins with FDA first-generic approval records, which define the launch events and approval dates used as the index point for diffusion measurement. These records are cleaned, filtered for study inclusion, and deduplicated so that each generic-brand launch is represented by its earliest approval date.

In parallel, Medicaid State Drug Utilization Data from 2017 through 2025 are staged and consolidated into a unified utilization table. These records provide the state-quarter prescription utilization data needed to observe whether generic use appears after approval.

The cleaned FDA launch table is then linked to Medicaid utilization records through a reproducible drug-name mapping process. Because public FDA and Medicaid datasets may use different naming conventions, the mapping step standardizes generic and brand-name terms before calculating utilization metrics.

After mapping, the SQL Server workflow constructs quarterly launch-level diffusion measures. For each launch and each quarter after approval, the system calculates active state-level molecule markets, generic-observed states, Numeric Distribution, Weighted Distribution, Access Gap, generic prescription volume, and total molecule prescription volume.

The first three quarters after approval are then transformed into a model-training dataset. The fourth-quarter Weighted Distribution is used to assign the first-year adoption outcome: Fast, Medium, or Slow. The resulting model-ready CSV files are exported from SQL Server and used by the ML.NET workflow.

The ML.NET workflow loads the train and test CSV files, constructs the numeric feature vector, normalizes features, trains a multiclass SDCA Maximum Entropy model, evaluates predictions using year-based validation, and exports the trained model artifact. Final SQL result tables and ML.NET validation metrics are then compiled into the study workbook for reproducibility.

This workflow creates a complete path from public source data to interpretable diffusion indicators and predictive validation outputs. A detailed list of SQL tables, scripts, and generated artifacts is provided in the reproducibility section.

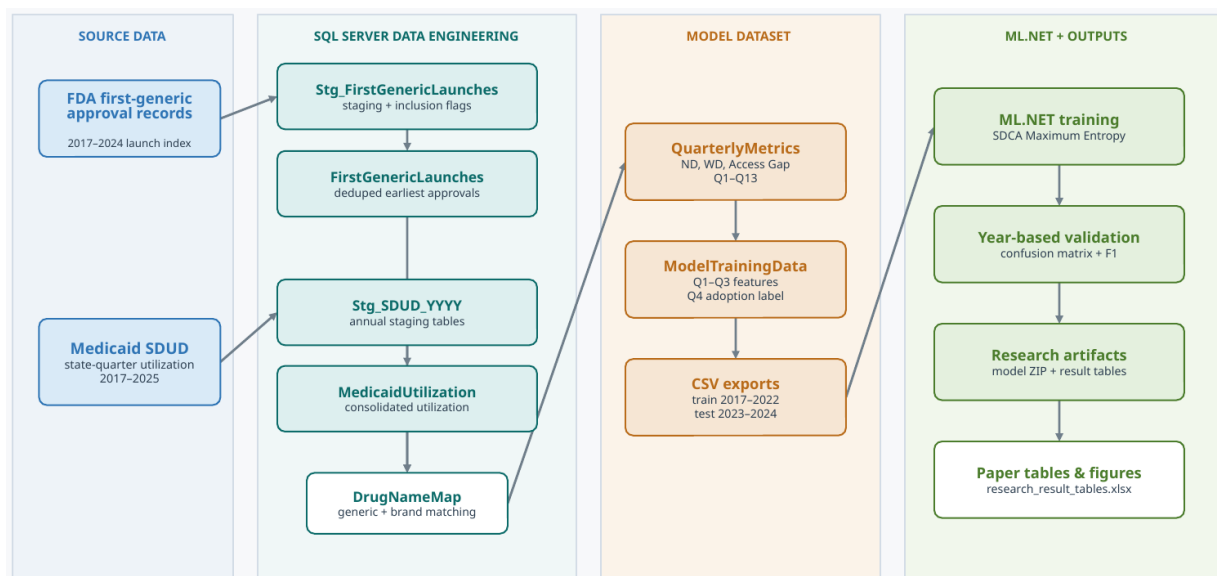


Figure 1. End-to-end SQL Server and ML.NET workflow from public FDA/CMS data to diffusion metrics, year-based validation, and paper-ready result tables.

4. Data Sources

4.1 FDA First Generic Approval Records

The FDA first-generic approval records were used to identify launch events [1,2]. Each included launch was represented by a generic ingredient, brand reference, approval date, and therapeutic class. The approval date was treated as the index date for measuring post-approval Medicaid diffusion.

The study period included first generic approvals from 2017 through 2024. After deduplication, 278 first-generic launch records were retained.

4.2 Medicaid State Drug Utilization Data

Medicaid State Drug Utilization Data were used to observe state-quarter prescription utilization [4,5]. The study used SDUD files from 2017 through 2025 to allow post-approval follow-up for 2017–2024 launch records. The state-quarter structure of SDUD makes it suitable for measuring geographic diffusion over time.

The analysis focuses on Medicaid utilization and should not be interpreted as a full U.S. market analysis. It does not capture commercial insurance, Medicare Part D, cash-pay transactions, hospital-administered products outside the dataset, or patient-level access.

4.3 Unit of Analysis

The main unit of analysis is a launch-quarter observation. A launch is defined as a generic-brand pair indexed by FDA approval date. For each launch, the pipeline computes quarterly diffusion metrics beginning with the first quarter after approval. The main adoption label is based on fourth-quarter Weighted Distribution. Model features are drawn from the first three quarters.

5. Measurement Design

5.1 Active State-Level Molecule Markets

For each launch and quarter, the pipeline first identifies active state-level molecule markets. A state is considered active for a molecule if total molecule utilization is observed in that state-quarter. This denominator is important because a state without any observed molecule utilization should not be counted as a failure of generic diffusion.

5.2 Numeric Distribution

Numeric Distribution measures the geographic breadth of generic utilization across active state-level Medicaid molecule markets. For each launch and quarter, it is calculated as:

$$ND = 100 \times \text{Generic-observed states} / \text{Active molecule states}$$

where **active molecule states** are state-quarter markets with observed Medicaid utilization for the molecule, and **generic-observed states** are active molecule states where utilization of the generic product is observed.

A higher Numeric Distribution means that generic utilization is observed across a larger share of active state-level Medicaid markets. Numeric Distribution is therefore a geographic breadth measure.

5.3 Weighted Distribution

Weighted Distribution measures the volume-weighted reach of generic utilization. For each launch and quarter, it is calculated as:

WD = $100 \times \text{Molecule prescription volume in generic-observed states} / \text{Molecule prescription volume in all active molecule states}$

This measure gives more weight to states with larger molecule-level prescription volume. Weighted Distribution can therefore be high even when Numeric Distribution is low. This occurs when generic utilization appears in a small number of high-volume states before becoming geographically broad.

5.4 Access Gap

Access Gap measures the difference between volume-weighted reach and geographic breadth. It is calculated as:

Access Gap = WD – ND

A large positive Access Gap indicates high-volume-first diffusion. In such cases, generic utilization is concentrated in states that represent a large share of molecule prescription volume, even though the generic has not yet appeared broadly across many state-level Medicaid markets.

5.5 Adoption Class Definition

The primary adoption label was defined using fourth-quarter Weighted Distribution:

- **Fast:** Q4 WD $\geq 80\%$
- **Medium:** Q4 WD $\geq 50\%$ and $< 80\%$
- **Slow:** Q4 WD $< 50\%$

The Q4 endpoint was selected to represent first-year post-approval diffusion. The model uses Q1–Q3 features to predict the Q4 label, creating a practical early-warning framework.

6. Machine Learning Workflow

6.1 Prediction Task

The prediction task is multiclass classification. The target variable is Q4 adoption class: Fast, Medium, or Slow. The model uses early diffusion features from Q1, Q2, and Q3.

Each launch contributes up to three model rows, one for each early quarter. The label is the same for each of those rows because the label is based on the launch's Q4 adoption class.

6.2 Features

The ML.NET model uses only numeric diffusion and utilization features:

- Quarter since approval
- Current Numeric Distribution
- Current Weighted Distribution

- Previous-quarter Numeric Distribution
- Previous-quarter Weighted Distribution
- Numeric Distribution growth
- Weighted Distribution growth
- Access Gap
- Total generic prescriptions
- Total molecule prescriptions

Generic name, brand name, and therapeutic class are not used as model features. They are retained in SQL tables and Excel result sheets for auditability, but excluded from the ML feature vector. This design reduces the risk that the model simply memorizes drug identities.

6.3 Train-Test Design

Two evaluation strategies are possible: random row-level validation and year-based validation. This paper emphasizes year-based validation because it is more appropriate for forecasting.

The year-based split was:

- Training set: launches from 2017–2022
- Test set: launches from 2023–2024

This split asks whether historical launch diffusion patterns can predict newer launch outcomes. It is more realistic than randomly mixing rows from the same time period.

6.4 ML.NET Implementation

The model was implemented in C# using ML.NET. ML.NET has been described as Microsoft’s machine-learning framework for .NET-based model development [10]. The pipeline maps the adoption class into a multiclass label, concatenates numeric features into a feature vector, applies min-max normalization, trains a multiclass SDCA Maximum Entropy classifier, maps predicted labels back to class names, and evaluates predictions using standard multiclass metrics [11,12].

The trained model is saved as a .zip artifact for reproducibility.

6.5 Evaluation Metrics

The evaluation reports:

- Micro accuracy
- Macro accuracy
- Log loss
- Confusion matrix
- Per-class precision
- Per-class recall

- Per-class F1-score
- Per-class support

Micro accuracy captures overall row-level performance. Macro accuracy is more sensitive to minority-class failure. Per-class metrics are essential because the Medium class is rare.

7. Results

7.1 Dataset Summary

The final analysis dataset contained 278 first-generic launch records after deduplication. The SQL Server pipeline produced 3,202 quarterly diffusion observations across quarters 1–13 after approval. The first four quarters contained 1,064 observations. A total of 265 launch cases had complete Q1–Q4 observations and were used for the main adoption-class analysis. The Q1–Q3 ML training table contained 795 rows.

Table 1. Dataset summary

Metric	Value
FDA launch records included after dedupe	278
Quarterly ND/WD observations Q1–Q13	3,202
First-four-quarter observations Q1–Q4	1,064
Complete Q1–Q4 launch cases	265
ML.NET training rows Q1–Q3	795

7.2 Launch Records by Approval Year

The included launch records covered FDA approval years 2017 through 2024.

Table 2. Included launch records by approval year

Approval year	Launch count
2017	38
2018	41
2019	52
2020	43
2021	26
2022	27
2023	22
2024	29

The distribution is not uniform across years, which reflects both the underlying FDA first-generic approval records and the inclusion/matching process used in the study.

7.3 Quarterly Metric Coverage

Quarterly coverage declined over time because later quarters require longer follow-up. Q1 and Q2 each had 267 distinct launches with observed metrics. Q3 through Q5 each had 265. By Q13, 214 launches remained observable.

Table 3. Quarterly metric coverage

Quarter since approval	Metric rows	Distinct launches
1	267	267
2	267	267
3	265	265
4	265	265
5	265	265
6	257	257
7	247	247
8	242	242
9	237	237
10	233	233
11	225	225
12	218	218
13	214	214

This coverage pattern supports the use of Q4 as the primary first-year adoption endpoint. Q4 has complete coverage for 265 launch cases, while later quarters naturally have lower availability.

7.4 Adoption Class Distribution

Among the 265 complete Q1–Q4 launch cases, 202 launches were classified as Fast, 6 as Medium, and 57 as Slow.

Table 4. Launch-level adoption class distribution

Adoption class	Launch count	Percent of launches
Fast	202	76.23%
Medium	6	2.26%
Slow	57	21.51%

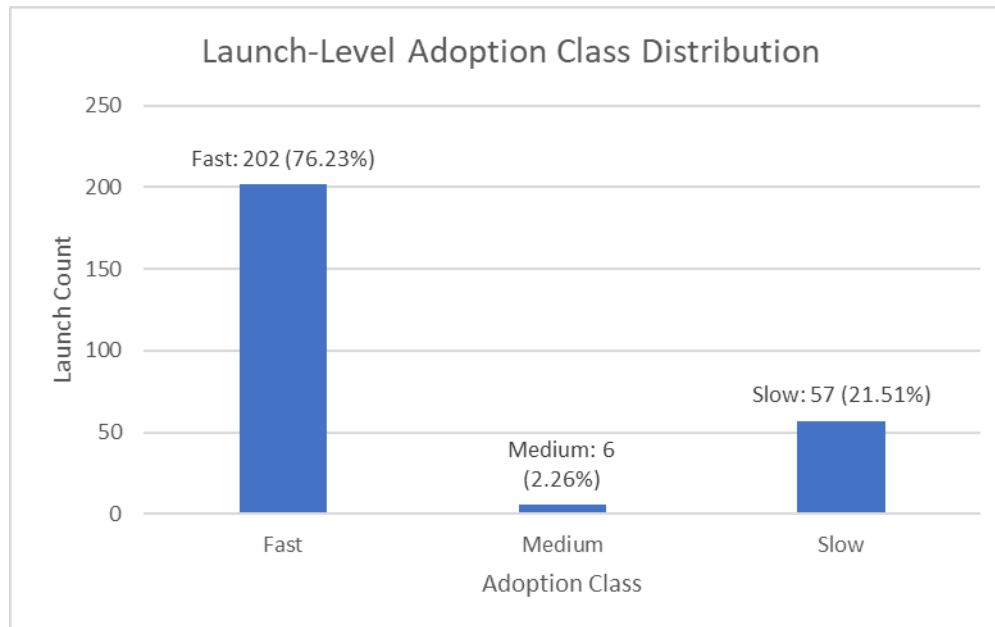


Figure 2. Launch-level adoption class distribution among complete Q1–Q4 first generic launch cases. Most launches reached Fast adoption by the fourth quarter, while Medium adoption was rare.

The main empirical pattern is clear: most first generic launches with complete first-year observations achieved broad volume-weighted Medicaid diffusion by Q4, but a substantial minority remained slow. Medium adoption was uncommon.

The ML training rows show the same class proportions because each complete launch contributes Q1–Q3 rows.

Table 5. ML training class distribution

Adoption class	Training rows	Percent of training rows
Fast	606	76.23%
Medium	18	2.26%
Slow	171	21.51%

This severe class imbalance is important for interpreting the machine learning results. The Medium class is too small to support strong three-class learning.

7.5 Time to Broad Weighted Distribution

A time-to-threshold analysis examined when launches first reached $WD \geq 80$. This analysis used the observed Q1–Q4 metric window and included 267 launch records with at least one early-quarter observation. The denominator differs slightly from the complete Q1–Q4 adoption-class analysis because the time-to-threshold analysis includes launches with at least one observed early-quarter metric, while the adoption-class analysis requires complete Q1–Q4 observations.

Table 6. Time to WD80 threshold

Timing group	Launch count	Percent of launches
Reached WD80 in Q1	160	59.93%
Reached WD80 in Q2	27	10.11%
Reached WD80 in Q3	13	4.87%
Reached WD80 in Q4	5	1.87%
Did not reach WD80 by Q4	62	23.22%

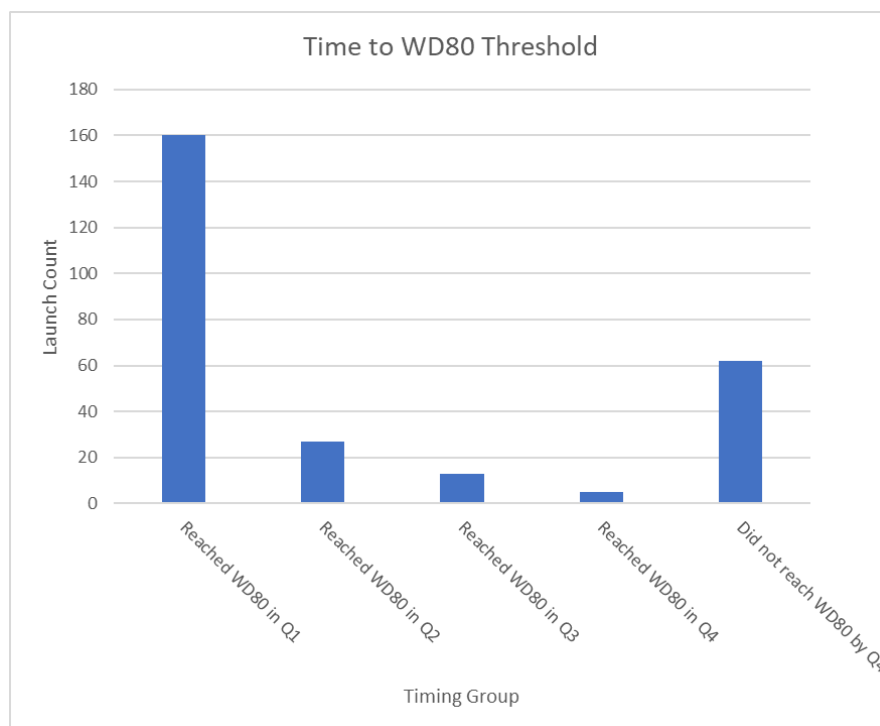


Figure 3. Time to reach WD80 among observed first generic launch cases. WD80 indicates Weighted Distribution of at least 80%.

The result shows that broad weighted diffusion, when it occurs, often occurs early. Nearly 60% of observed early-window launches reached WD80 by Q1. However, 23.22% did not reach WD80 by Q4, consistent with the Slow-adoption group observed in the adoption-class table.

7.6 High Access Gap Cases

The Access Gap measure identifies launches where volume-weighted reach substantially exceeds geographic breadth. These are cases where generic utilization appears in states representing large prescription volume but not necessarily in many states.

Table 7. Top Access Gap cases in Q1–Q4

Generic name	Brand name	Therapeutic class	Year	Quarter since approval	ND %	WD %	Access Gap
Droxidopa	Northera	CNS / Cardiovascular	2021	1	16.67	84.30	67.63
Icatibant	Firazyr	Rare Disease / Allergy	2019	1	16.67	80.90	64.24
Brinzolamide	Azopt	Ophthalmology	2021	2	8.57	71.29	62.72
Tasimelteon	Hetlioz	CNS / Sleep Disorder	2023	2	14.29	76.29	62.00
Posaconazole	Noxafil	Infectious Disease / Antifungal	2019	1	20.83	82.64	61.80
Aliskiren	Tekturna	Cardiovascular	2019	1	9.09	68.80	59.71
Etravirine	Intelence	HIV / Antiviral	2021	1	7.69	67.26	59.57
Ambrisentan	Letairis	Cardiovascular / Pulmonary Hypertension	2019	2	11.90	71.43	59.53
Pazopanib	Votrient	Oncology	2024	2	20.00	78.88	58.88
Maraviroc	Selzentry	HIV / Antiviral	2022	1	17.65	74.99	57.34

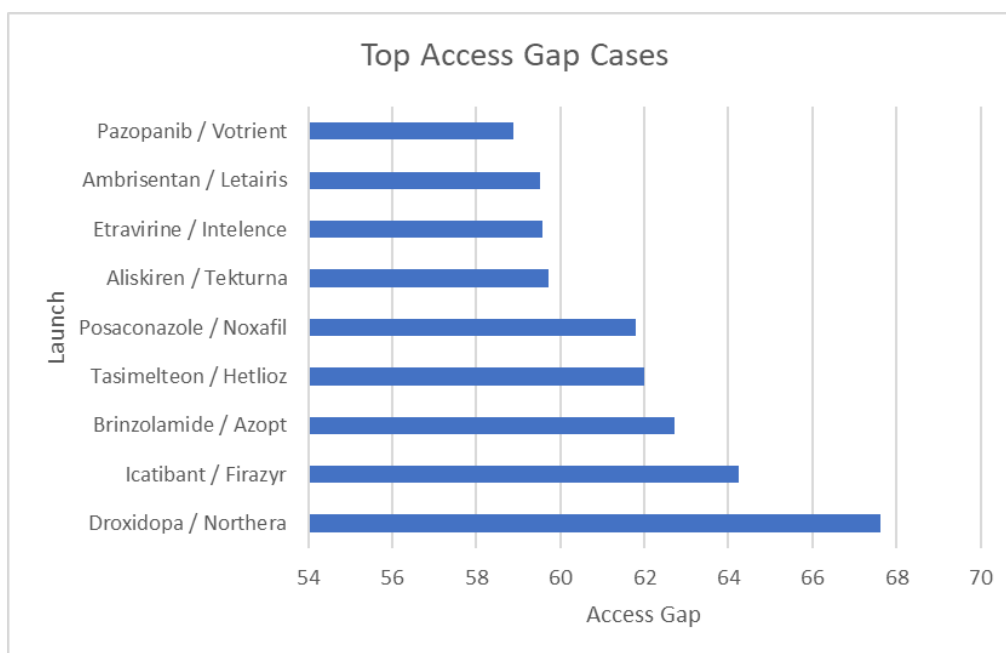


Figure 4. Top Access Gap cases in the first four quarters after approval. Larger values indicate high-volume-first diffusion, where generic utilization appears in states representing large prescription volume before becoming geographically broad.

These examples demonstrate why ND and WD should be reported together. A launch can have low state-level breadth but high volume-weighted reach. Such a launch is not simply “low adoption”; it is concentrated adoption. From a market-access perspective, this indicates that large states may receive or record generic utilization earlier than smaller markets.

7.7 Slow Q4 Cases

Some launches showed no observed generic utilization by Q4 despite active molecule-level utilization. Examples include Cinacalcet Hydrochloride/Sensipar, Eslicarbazepine Acetate/Aptiom, Darunavir/Prezista, Nilotinib/Tasigna, Ertugliflozin/Steglatro, and Apremilast/Otezla. These cases had Q4 ND and WD of zero in the result table.

Such cases should be interpreted carefully. A zero observed generic value in Medicaid SDUD does not necessarily mean the generic was unavailable nationally. It means that the pipeline did not observe generic utilization in the relevant state-level Medicaid molecule markets during the measured quarter. Possible explanations include delayed commercialization, limited Medicaid uptake, mapping limitations, low volume, payer coverage dynamics, or product-specific market factors.

7.8 Therapeutic Class Patterns

Therapeutic class summaries showed variation in Q4 diffusion. Some classes, such as Dermatology and Gastrointestinal, had high mean Q4 WD and many Fast launches. Other classes, such as Oncology, Diabetes, Rare Disease/Specialty, and CNS/Multiple Sclerosis, showed lower mean Q4 WD and more Slow cases.

These class-level summaries are descriptive rather than causal. They suggest that diffusion may differ across therapeutic areas, but the study is not designed to estimate why those differences occur.

7.9 Threshold Sensitivity

Because the adoption labels depend on Q4 WD thresholds, a sensitivity analysis was performed.

Table 8. Threshold sensitivity

Threshold scheme	Fast count	Medium count	Slow count
Scheme A: Fast ≥ 80 , Medium 50–79.99, Slow < 50	202	6	57
Scheme B: Fast ≥ 75 , Medium 40–74.99, Slow < 40	205	3	57
Scheme C: Fast ≥ 90 , Medium 60–89.99, Slow < 60	188	18	59

The Slow class is relatively stable across the tested threshold schemes, ranging from 57 to 59 launches. The Medium class is more sensitive to threshold selection. This supports a cautious interpretation: the most robust distinction is between broadly diffused launches and launches that remain below broad diffusion by Q4. The Medium class is best understood as a threshold-dependent transition zone.

7.10 Data Quality Checks

The study pipeline included data quality checks for duplicate metric rows and null model fields.

Table 9. Data quality checks

Check	Issue count
Duplicate metric rows	0
Null CurrentND in ModelTrainingData	0
Null CurrentWD in ModelTrainingData	0
Null AdoptionClass in ModelTrainingData	0
Null ApprovalYear in ModelTrainingData	0

These checks support the internal consistency of the final result tables and the year-based ML dataset.

8. ML.NET Validation Results

8.1 Year-Based Split

The year-based validation split trained the model on 2017–2022 launch observations and tested it on 2023–2024 launch observations.

Table 10. Year-based train-test split

Split group	Adoption class	Row count
Train: 2017–2022	Fast	522
Train: 2017–2022	Medium	9
Train: 2017–2022	Slow	123
Test: 2023–2024	Fast	84
Test: 2023–2024	Medium	9
Test: 2023–2024	Slow	48

The split shows the same core imbalance problem: Medium remains rare in both training and testing.

8.2 Overall Metrics

Table 11. ML.NET year-based validation performance

Metric	Value
Training years	2017–2022
Test years	2023–2024
Training rows	654
Test rows	141
Micro accuracy	90.78%
Macro accuracy	65.08%
Log loss	0.4734

The high micro accuracy indicates strong overall performance. The lower macro accuracy indicates that performance is uneven across classes.

8.3 Confusion Matrix

Year-Based Validation Confusion Matrix			
ActualClass	PredictedFast	PredictedMedium	PredictedSlow
Fast	80	0	4
Medium	9	0	0
Slow	0	0	48

Figure 5. Confusion matrix for the year-based ML.NET validation. The model correctly identified most Fast and Slow observations but did not predict the rare Medium class.

The model correctly classified 80 of 84 Fast rows and all 48 Slow rows. It did not predict any Medium rows.

8.4 Per-Class Performance

Table 12. Per-class validation metrics

Class	Precision	Recall	F1	Support
Fast	89.89%	95.24%	92.49%	84
Medium	0.00%	0.00%	0.00%	9
Slow	92.31%	100.00%	96.00%	48

These results should not be described as uniformly strong multiclass classification. The correct interpretation is that early diffusion features separate Fast and Slow outcomes well, while Medium adoption is not learned under the current data and threshold design. This is an important result, not merely a weakness. It shows that the observed diffusion process is dominated by broad adoption and slow adoption, with few stable intermediate cases.

9. Discussion

9.1 First Generic Approval Begins the Access Process; It Does Not Complete It

The results support the view that regulatory approval should be treated as the beginning of a diffusion process. Although many launches reached broad weighted Medicaid diffusion quickly, a substantial minority remained below the broad diffusion threshold by Q4. This means that approval-based counts alone cannot describe realized utilization.

This finding is important for market-access monitoring. A first generic may be approved, but payer-market utilization may still be delayed, absent, or geographically concentrated. Public data can reveal these patterns after approval.

9.2 ND and WD Measure Different Types of Diffusion

Numeric Distribution and Weighted Distribution provide complementary views. ND measures breadth across state-level markets. WD measures the prescription-volume importance of the states where generic use is observed. Access Gap connects the two by identifying imbalanced diffusion.

The top Access Gap cases show that some generics appear in a small number of states that account for a large share of molecule volume. This high-volume-first pattern may be commercially meaningful but geographically uneven. Without both ND and WD, such cases could be misunderstood.

9.3 Early Diffusion Metrics Can Identify Slow Adoption Risk

The ML.NET model performed especially well on Slow cases in the year-based test set. It achieved 100% recall and 96.00% F1 for Slow observations. This suggests that early post-approval diffusion metrics may be useful as an early-warning signal for launches that are unlikely to achieve broad weighted adoption by Q4.

This does not mean the model is ready for operational deployment. It means that the measurement framework captures predictive information. Future work can improve the model by expanding data coverage, refining launch-date definitions, improving drug-name matching, and testing binary or time-to-event models.

9.4 The Medium Class Is Not Yet a Stable Modeling Target

The model failed to predict Medium cases. This is mainly explained by class imbalance: only 6 launch-level cases and 18 row-level observations were Medium in the main dataset, and the year-based training set contained only 9 Medium rows.

The threshold sensitivity analysis supports this interpretation. The number of Medium cases changes substantially under alternate threshold schemes. Therefore, Medium should be treated cautiously. The main robust finding is not that there are three equally meaningful adoption groups, but that early diffusion features distinguish broad adoption from weak adoption.

9.5 SQL Server and ML.NET Provide a Reproducible Public-Data Workflow

The workflow design is a major contribution. SQL Server handles reproducible staging, cleaning, joining, metric construction, and result-table generation. ML.NET handles model training, validation, and artifact export. The separation between data engineering and model training makes the pipeline transparent and auditable.

This is important because public administrative data analysis can easily become opaque. By preserving intermediate tables such as FirstGenericLaunches, MedicaidUtilization, QuarterlyMetrics, and ModelTrainingData, the framework allows future users to inspect each transformation stage.

10. Limitations

This study has several limitations.

First, FDA approval date is used as the index date. Approval date is not always the same as commercial launch date, supply availability date, or payer coverage date. Some products may be approved but not immediately marketed.

Second, Medicaid SDUD captures Medicaid outpatient drug utilization and does not represent the full U.S. prescription market. Results may differ in Medicare, commercial insurance, cash-pay, hospital, or specialty distribution channels.

Third, the analysis is state-quarter level. It does not measure patient-level access, patient cost, pharmacy stocking, formulary placement, prior authorization, prescriber behavior, or substitution decisions.

Fourth, drug-name matching between FDA approval records and Medicaid utilization data is imperfect. Public datasets may use different names, dosage-form descriptions, strengths, labeler records, and product naming conventions. Some false matches or missed matches may remain.

Fifth, therapeutic class labels were manually standardized for interpretability. This improves readability but introduces possible subjectivity.

Sixth, the adoption classes are threshold-based. The sensitivity analysis shows that Medium is especially threshold-dependent.

Seventh, the ML dataset is modest in launch-level size. Although the row-level model table contains 795 observations, these come from 265 complete launches. The Medium class is too small for reliable multiclass learning.

Eighth, the study is descriptive and predictive, not causal. It does not estimate the causal effect of generic approval on utilization or identify the causal drivers of slow diffusion.

Ninth, the model uses row-level Q1–Q3 observations. Multiple rows from the same launch are related, which means future work should also test launch-level aggregation, grouped validation, and time-to-event methods.

11. Future Work

Future work should extend the framework in five directions. First, future versions should incorporate commercial launch dates, supply availability information, or market-entry dates when available, because FDA approval date may precede actual market availability. Second, drug-name matching can be improved using stronger normalization, NDC-level linkage, dosage-form matching, and validation against FDA Orange Book or product-level reference data [3]. Third, the predictive task can be reformulated as a binary or time-to-event problem to better handle the rare Medium class. Fourth, additional payer datasets, including Medicare Part D or commercial claims, could be used to compare Medicaid diffusion with broader U.S. market adoption. Finally, future work should investigate therapeutic-class-specific and policy-level factors that may explain why some first generics diffuse rapidly while others remain slow or geographically concentrated.

12. Reproducibility and Code Availability

This study was designed as a reproducible public-data workflow. The SQL Server pipeline, ML.NET training code, model-ready CSV files, validation outputs, and result workbook are maintained in a public GitHub repository. The repository includes the SQL scripts used to stage FDA first-generic approval records, process Medicaid State Drug Utilization Data, calculate quarterly diffusion metrics, create the model-training dataset, and export final result tables.

The main reproducibility artifacts include the cleaned launch table, consolidated Medicaid utilization table, quarterly diffusion metrics, model-training CSV files, ML.NET validation code, trained model artifact, and Excel result workbook. These artifacts allow the study workflow to be reviewed from source data processing through final model evaluation.

The public repository and archived code/data record are available for review and reuse [13,14].

13. Conclusion

This study develops a reproducible SQL Server and ML.NET framework for measuring and forecasting post-approval diffusion of first generic drugs in state-level Medicaid pharmacy markets. The framework links FDA first-generic approval records with Medicaid State Drug Utilization Data, computes quarterly Numeric Distribution, Weighted Distribution, and Access Gap metrics, and uses early diffusion signals to predict first-year adoption class.

The results show that most complete first-year launches achieved broad weighted Medicaid diffusion by Q4, but a substantial minority remained slow. High Access Gap cases demonstrate that volume-weighted reach and geographic breadth can diverge substantially. The ML.NET model predicted Fast and Slow adoption patterns well in year-based validation but failed to identify the rare Medium class, highlighting both the usefulness and the limits of the current multiclass design.

The main contribution is not only a predictive model. It is an auditable measurement framework that turns public regulatory and utilization data into interpretable post-approval diffusion indicators. This framework can support future research on generic access, payer-market uptake, geographic diffusion, and early identification of slow adoption patterns.

Conflict of Interest / Disclosure

The author declares no conflict of interest. This research was conducted independently using publicly available FDA and Medicaid datasets. The analysis, interpretations, and conclusions are the author's own.

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